

Objective: Ultracentrifugation in the WX Ultra Series can support only ~30ml samples at a time, and therefore do not significantly concentrate viruses. Tangential flow filtration enables viral concentration of large volumes that can then be purified using sucrose density gradients. **It is imperative that the system be cleaned properly between use in order to prevent cross-contamination. You may also want to determine that the cleaning solution used will actually inactivate/degrade/kill whatever you are concentrating.**

Materials

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| -Millipore Labscale TFF System | -50mM Tris-HCl, pH=7.8, 10mM MgCl ₂ |
| -Pellicon XL 30kDa filter (Cat No. PXC030C50) | -Milli-Q Water |
| -Filtered viral lysate; titer known. | -Manual, tubing, 10ml syringe, TFF parts. |

Machine Preparation

- 1) Set up TFF apparatus so that the filter is attached. Both the retentate and waste tubes should empty into your waste flask (Figure 1).
- 2) Make 500mls of 0.1N NaOH, 45°C. Pour into the TFF reservoir.
- 3) Turn the pump speed to 1 or 2, and gradually turn the back pressure valve clockwise to increase the pressure. The bottom pressure gauge should be 30-40 psi; top pressure gauge should be ~10 psi.
- 4) Circulate the fluids until ~350mls are left. Re-attach the retentate tube to circulate the remaining cleaning solution that doesn't pass through the filter. Run the NaOH solution through system for ~20 minutes (Don't let the system run dry!).
- 5) When 50mls are left, turn the pump speed to off, and tighten a plastic clamp onto the tubing that flows to the pump inlet. Detach the tube, and undo the clamp to empty the remaining NaOH solution into the waste vial (Figure 2).
- 6) Next, attach a syringe to the tube connecting to the pump outlet, and pull back the plunger to draw out any remaining NaOH inside the machine. Re-attach the tubing, and use the syringe to drain any other tubing in the same manner (e.g. over the surface of the filter).
- 7) Wash the TFF system with at least 1L of MilliQ water, allowing the first 500mls to empty into the flask through both the ret. and waste tubes. Re-attach the ret. tube to circulate the last 500 mls and run it through for 10 minutes. Empty the remaining volume in the same process outlined in steps 5 and 6.
- 8) Now, you will 'prime' the TFF system with buffer to protect your viruses against any residual effects of the NaOH. The manual calls for 50mls of this solution, but I used 300mls of 50mM Tris-HCl, pH=7.8, 10mM MgCl₂. Use whatever buffer is appropriate for your system.
- 9) When roughly ~25mls of buffer are left, fill the reservoir with your sample. Continue to run at the same psi (30-40 psi on the bottom pressure; ~10 psi on the top pressure gauge). After you have concentrated your sample, empty the remaining volume in the reservoir into a falcon tube. Virus may get stuck to the filter, so use a syringe attached to the pump outlet tube, and draw out remaining fluid from the filter. You may have to go back and forth with the plunger to dislodge viruses that are stuck to the surface.
- 10) Purify your samples (e.g. filter sterilize if axenic cultures, perform sucrose density gradient if not).
- 11) Titer the sample by plaque assay and determine your actual concentration compared to your theoretical. *Theoretical Concentration:* $(V1 \times C1) = (V2 \times C2)$
- 12) Clean the TFF system with NaOH, rinse with Milli-Q, and use a syringe filter fill the Pellicon filter with 0.1N NaOH, 4°C until the next use.

Diagnosing Issues and Troubleshooting

- 1) If the pressure suddenly decreases (and you haven't adjusted the pump speed or back pressure), then the pressure may have been too high and caused a tear in the filter. This would allow the fluid to flow freely through the filter and into the waste vial much quicker than normal. Prepare a sample from the waste vial for epifluorescent microscopy to confirm.
- 2) If your actual concentration is much lower than the theoretical concentration, then one of a few things may be to blame:
 - a. There may be a tear in the filter.
 - b. The viruses may be sticking to the filter – use a syringe to force fluid into a falcon tube.
 - c. The viruses may be getting inactivated from residual NaOH on the filter or in the TFF system. This is unlikely if you have flushed with sufficient Milli-Q and buffer prior to adding sample.
- 3) If your sample is concentrating very slowly (barely dripping), then you may need to check the pressure. If the back pressure is not ~10 psi, it will concentrate more slowly, because fluid is circulating more than it is passing through the filter. The back pressure also cannot increase if the pump speed is too slow.

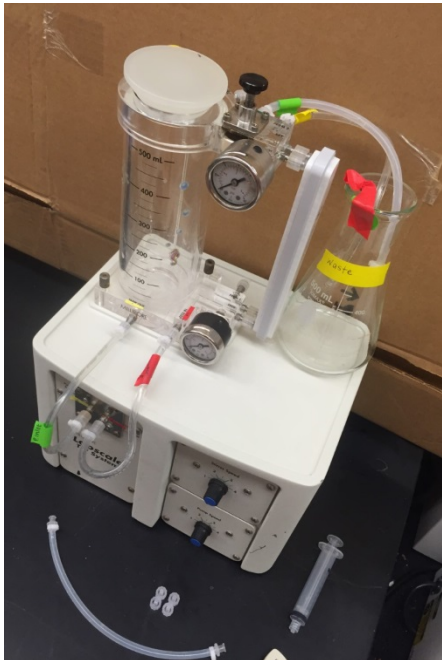


Figure 1: Apparatus set-up in step 1 of this protocol. The four pieces on the benchtop come from the filter ports, and the tubing is in place of the filter when not in use.



Figure 2: The clamp is tightened onto the P.inlet tube, and the tube has been detached from the P. Inlet port so that when the clamp is loosened, the volume in the reservoir can be emptied into another container